

MACHINE LEARNING AND BIO-INSPIRED OPTIMIZATION

ASSIGNMENT 1

1. Introduction

The Multi-Armed Bandit (MAB) problem is a fundamental concept in reinforcement learning. It presents the challenge of balancing exploitation—choosing the action with the highest estimated reward based on current knowledge—and exploration—selecting suboptimal actions to gather more information and potentially discover better long-term strategies. While exploitation maximizes immediate rewards, effective exploration can lead to greater cumulative rewards over time.

This project focuses on implementing and analysing different action selection strategies to address this trade-off. We examine the ϵ -greedy algorithm with varying values of ϵ , which determines the probability of exploration, as well as the Upper Confidence Bound (UCB) method. We also compare the performance of sample averaging vs. recency-weighted averaging for action-value estimation. By evaluating these strategies over multiple trials, we aim to assess their effectiveness in optimising long-term reward accumulation.

2. Implementation of Multi-Armed Bandit Algorithm (Problem 1)

2.1 Experiment 1: ϵ -Greedy with Sample Averaging Method

Setup:

- **ϵ values tested:** 0 (pure exploitation), 0.01, and 0.1.
- **Performance metrics:** Average reward and % optimal action selection.

Results Analysis:

ϵ Value	Observation on Average Reward	Observation on Optimal Action Selection
$\epsilon = 0$	Leads to suboptimal behaviour; agent gets stuck in a local optimum.	Low optimal action selection due to lack of exploration.
$\epsilon = 0.01$	Minimal exploration slightly improves learning but still struggles with local optima.	Moderate improvement in optimal action selection.
$\epsilon = 0.1$	More frequent exploration leads to better long-term learning.	Highest percentage of optimal action selection.

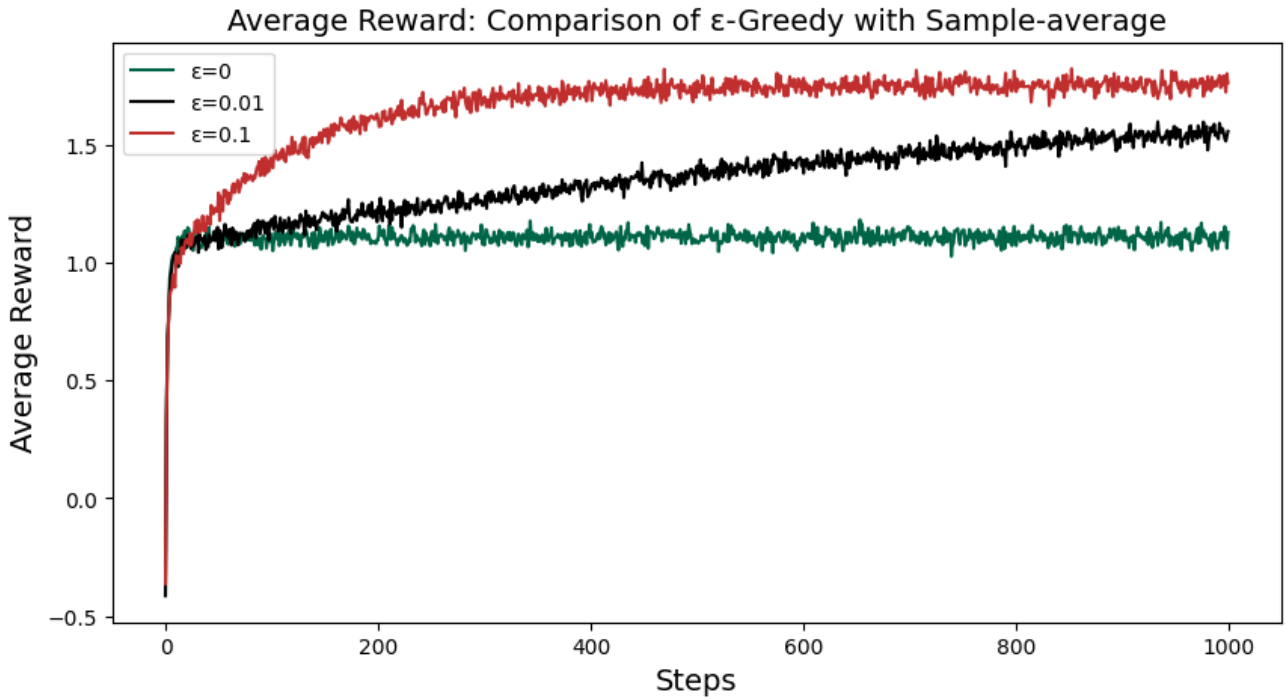


Figure 1: Average reward analysis for different ϵ values.

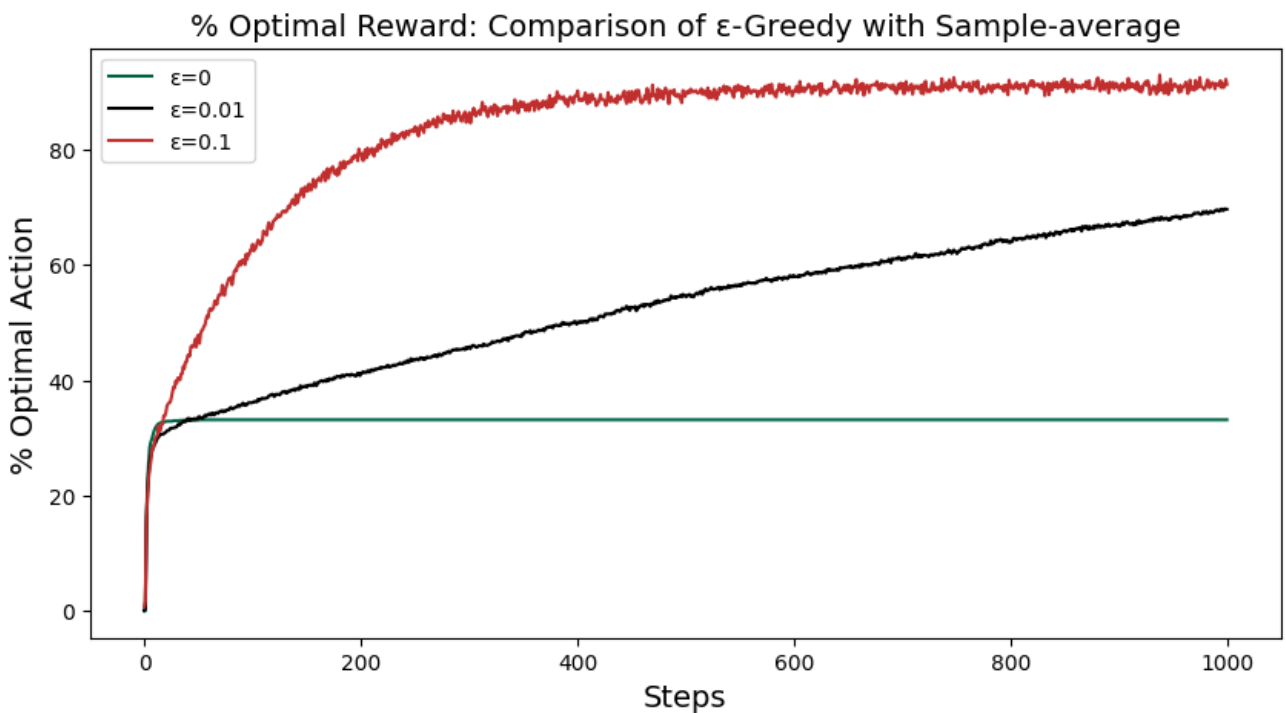


Figure 2: Optimal action selection for different ϵ values.

2.2 Experiment 2: Sample-Average vs. Exponential, Recency-Weighted Average (Stationary Machine, $\epsilon = 0.1$)

Setup:

- **Methods compared:** Sample-average vs. Exponential, recency-weighted average ($\alpha = 0.1$).

- **Stationary environment:** True action values remain constant.

Results Analysis:

- Sample-average initially outperforms exponential, recency-weighted average but both methods converge to similar performance over time.
- Exponential recency-weighted average does not provide significant advantages in a stationary environment.

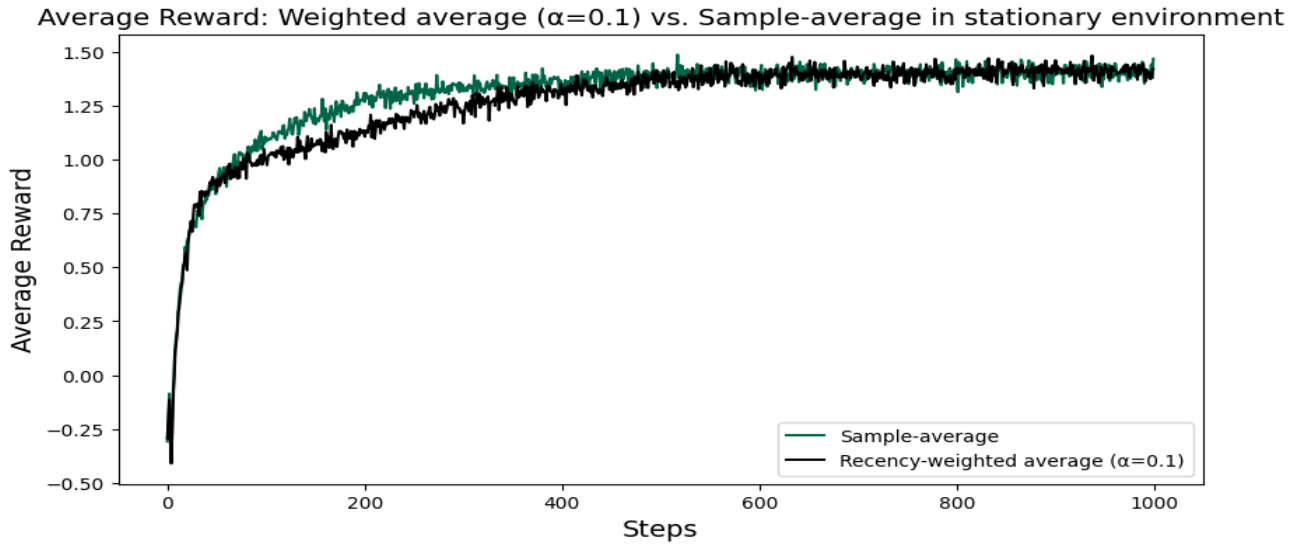


Figure 3: Comparison of average reward

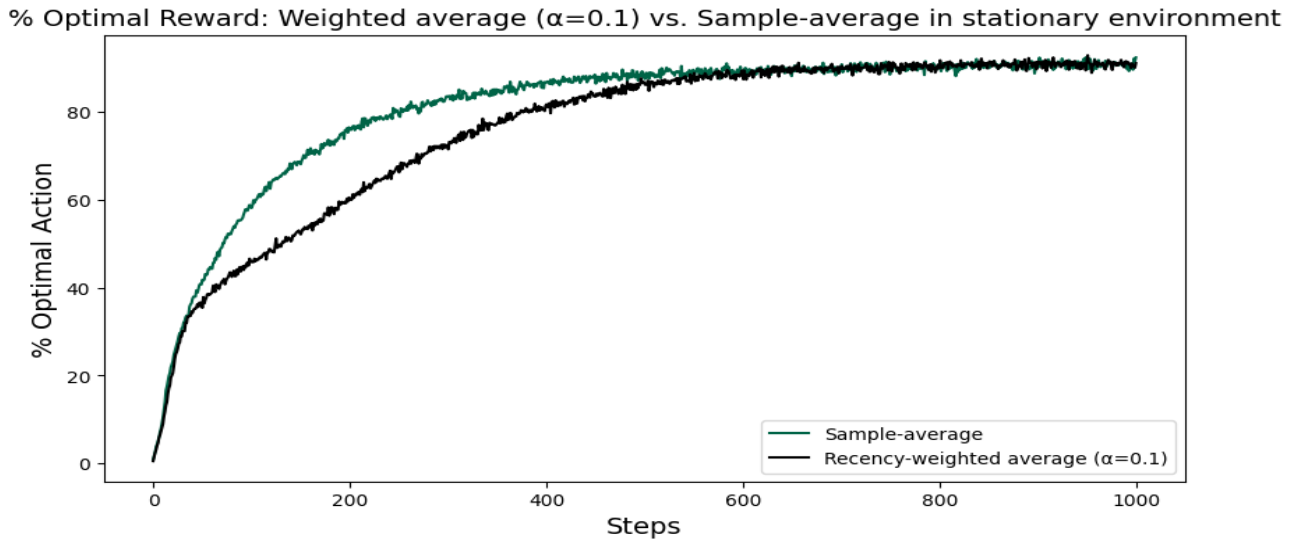


Figure 4: Comparison of optimal action selection.

2.3 Experiment 3: Sample-Average vs. Exponential, Recency-Weighted Average (Non-Stationary Machine, $\epsilon = 0.1$)

Setup:

- **Methods compared:** Sample-average vs. Exponential, recency-weighted average ($\alpha = 0.1$).

- **Non-stationary environment:** True action values drift over time.

Results Analysis:

- Exponential, recency-weighted average outperforms sample-average in a non-stationary setting.
- Weighted average method adapts better to changes in true action values over time.

Average Reward: Weighted average ($\alpha=0.1$) vs. Sample-average in nonstationary environment

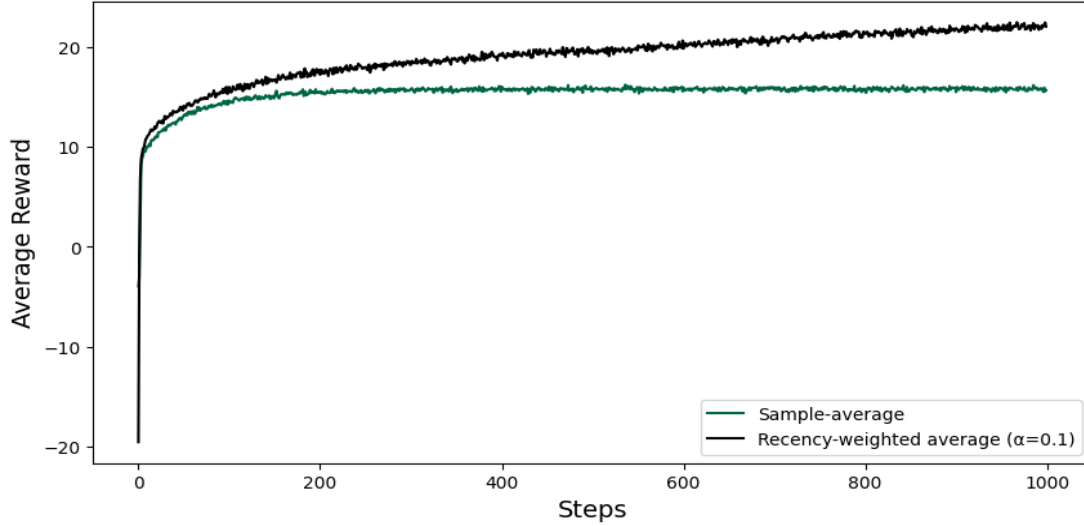


Figure 5: Average reward comparison for non-stationary environment.

% Optimal Reward: Weighted average ($\alpha=0.1$) vs. Sample-average in nonstationary environment

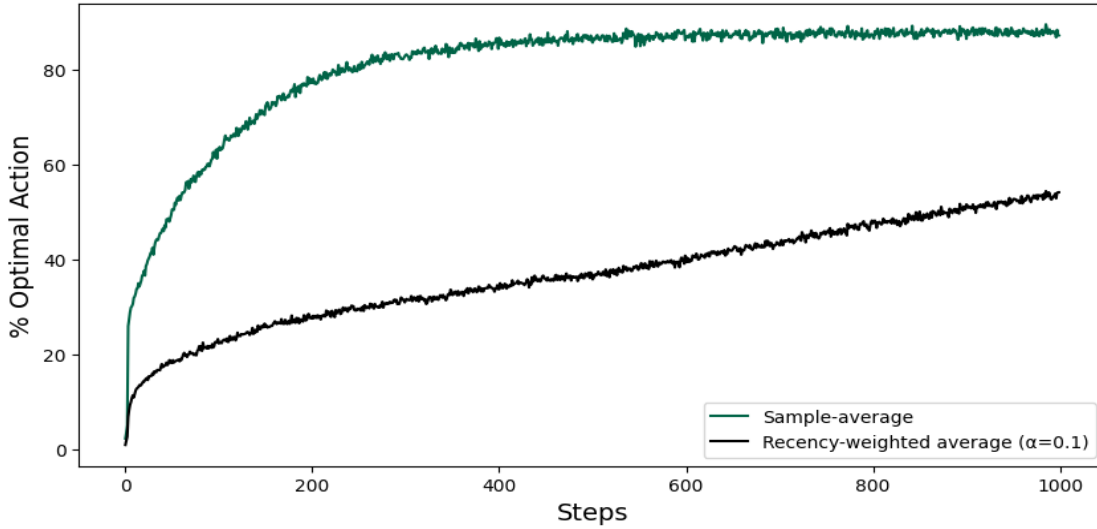


Figure 6: Optimal action selection for non-stationary environment.

2.4 Experiment 4: Optimistic ($Q=5$) vs. Realistic ($Q=0$) Initial Values

Setup:

- **Optimistic Initial Values:** $Q = 5$, $\epsilon = 0$ (greedy).
- **Realistic Initial Values:** $Q = 0$, $\epsilon = 0.1$ (ϵ -greedy).

Results Analysis:

- Realistic initial values perform better initially, but optimistic initial values outperform in the long run.

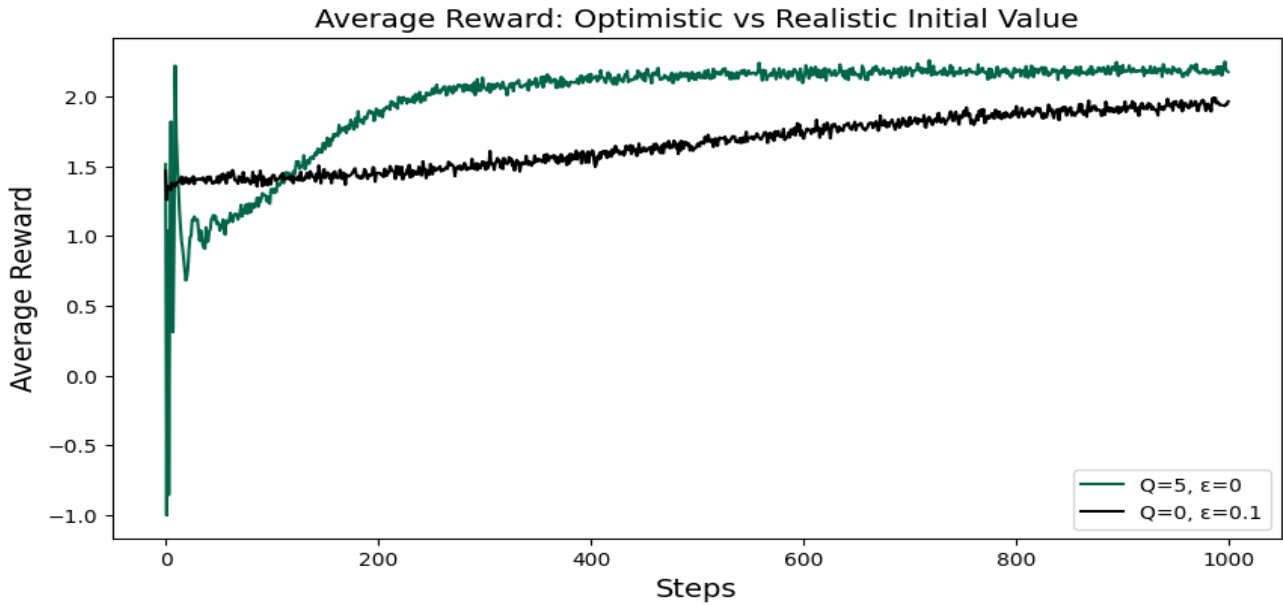


Figure 7: Average reward comparison for optimistic vs. realistic initialization.

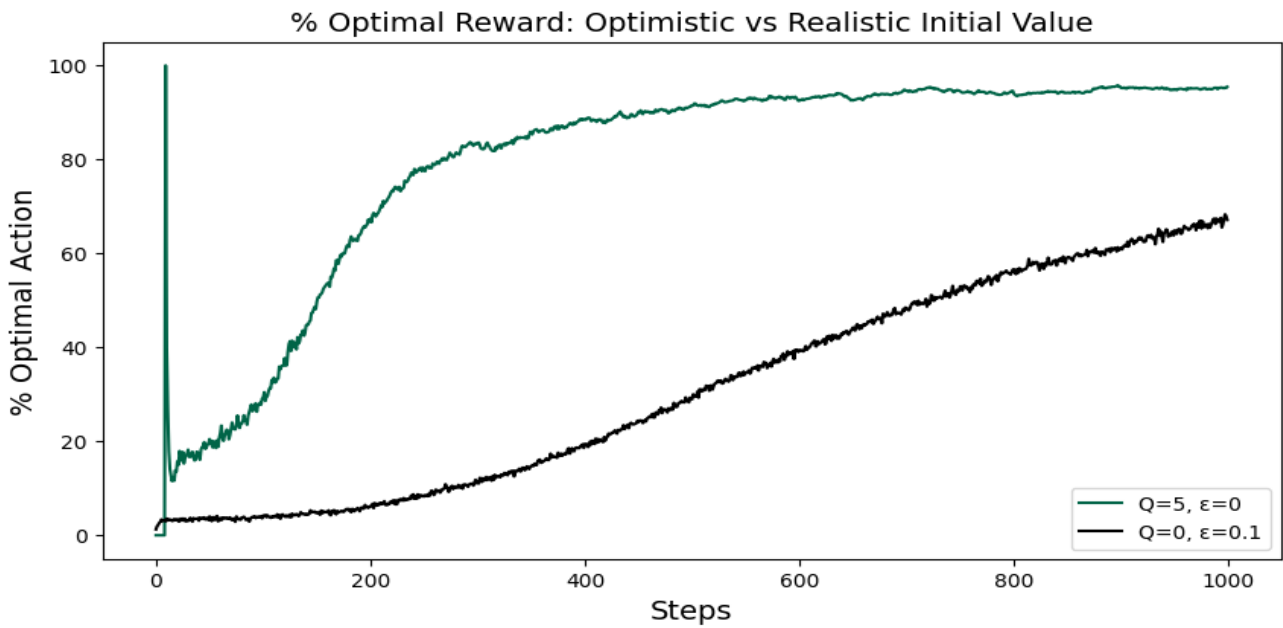


Figure 8: Optimal action selection comparison.

2.5 Experiment 5: UCB ($c=2$) vs. ϵ -Greedy ($\epsilon=0.1$)

Setup:

- Methods compared:** Upper Confidence Bound (UCB, $c=2$) vs. ϵ -Greedy ($\epsilon = 0.1$).

Results Analysis:

- UCB outperforms ϵ -Greedy in both early and later stages of training.

- UCB selects actions based on uncertainty, leading to better long-term performance.

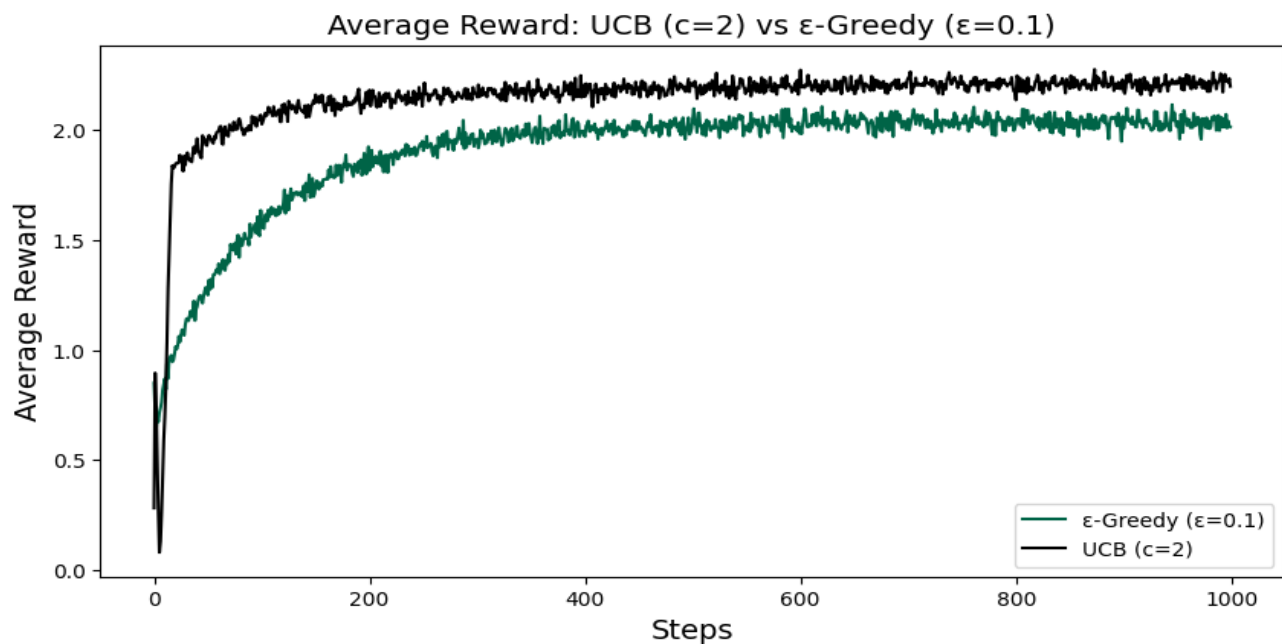


Figure 9: Comparison of UCB and ϵ -Greedy in average reward.

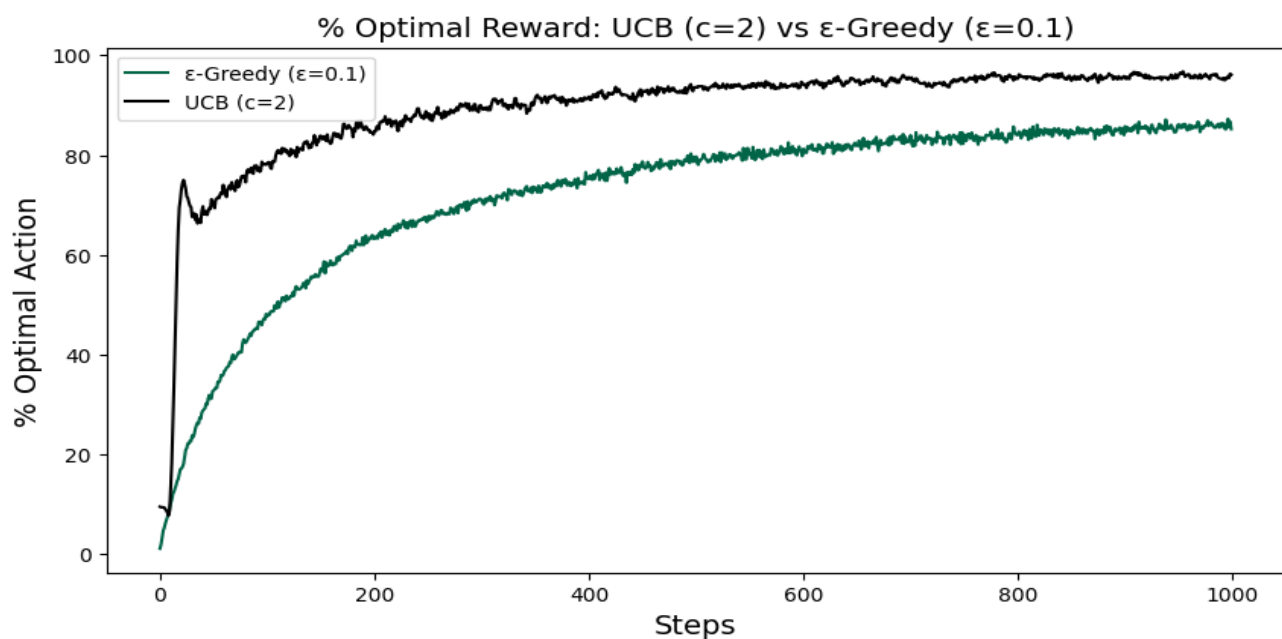


Figure 10: Comparison of UCB and ϵ -Greedy in optimal action selection.

3. Exploration and Exploitation (Problem 2)

Exploration and exploitation are fundamental concepts in reinforcement learning, forming a critical trade-off in decision-making.

3.1 Understanding Exploration and Exploitation

- Exploration: The process of discovering new actions and opportunities.
- Exploitation: Acting on the best-known actions to maximize rewards.

A reinforcement learning agent faces the exploration-exploitation dilemma, where it must balance between these two approaches. If the agent exploits only what it knows, it risks missing better strategies. Conversely, excessive exploration may lead to inefficient reward accumulation. The main challenge is to prefer actions that appear best while ensuring adequate exploration for trustworthy decision-making, particularly in stochastic environments.

3.2 Comparison of Pure Exploration vs. Pure Exploitation

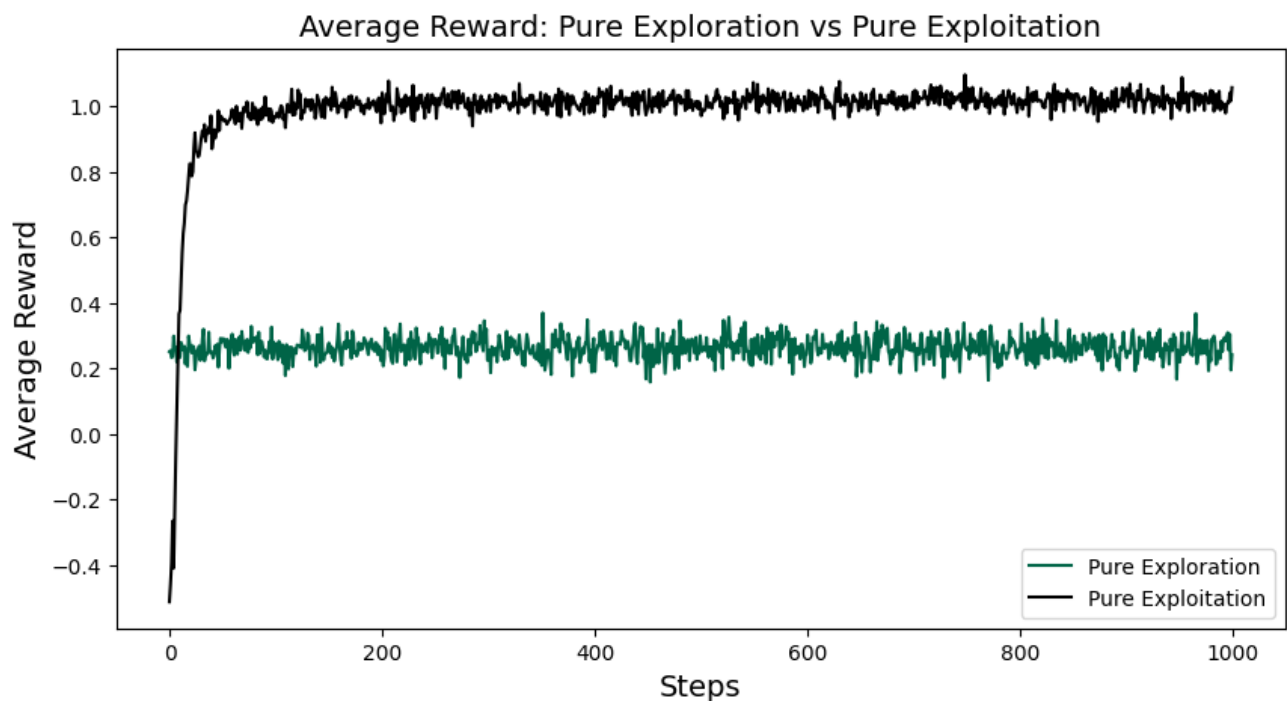


Figure 11: Comparison of pure exploration and pure exploitation in average rewards.

- Average Reward Graph (Figure 11):
 - Pure exploitation quickly converges to a higher average reward, indicating that the agent finds one good action early and continues exploiting it.
 - Pure exploration maintains a lower average reward since the agent randomly chooses actions and lacks consistency.

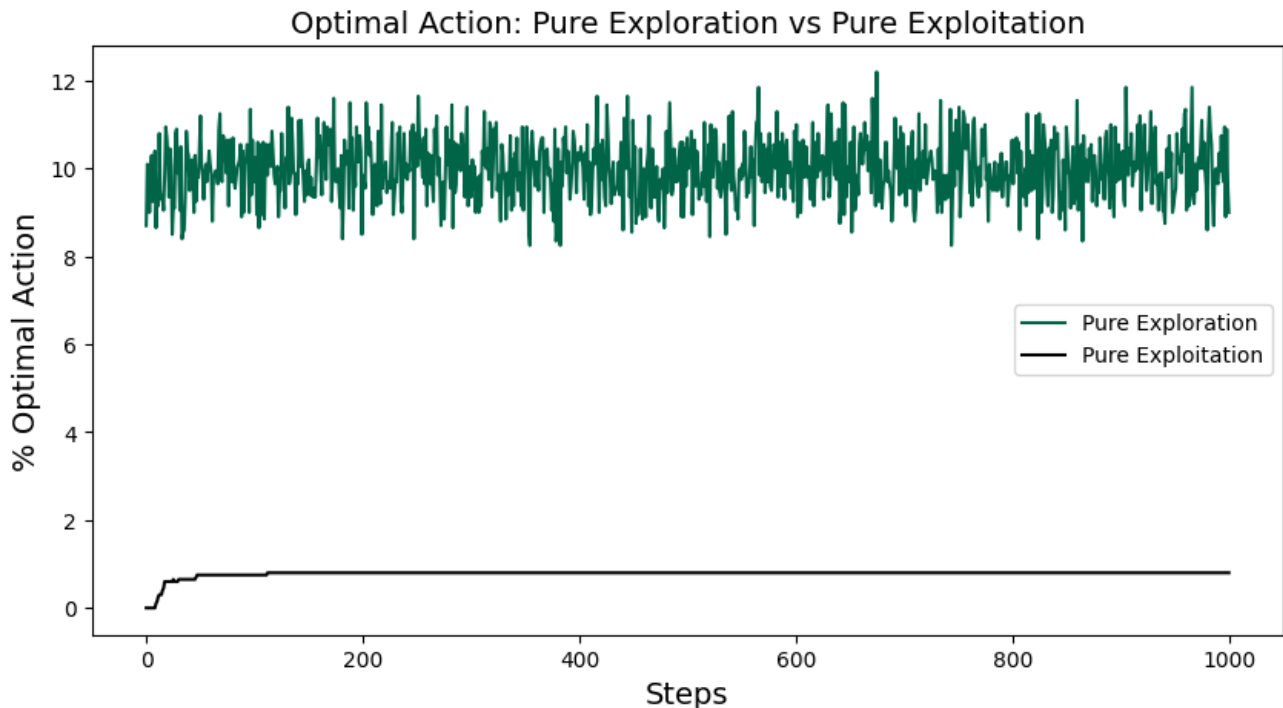


Figure 12: Comparison of pure exploration and pure exploitation in optimal action selection.

- **Optimal Action Selection Graph (Figure 12):**
 - In pure exploitation, the agent locks onto an action it believes is optimal and ceases exploration, limiting long-term performance.
 - In pure exploration, the agent randomly selects actions, increasing the likelihood of selecting the optimal action multiple times.

3.3 Combination of Exploration and Exploitation

A trade-off between exploration and exploitation is necessary, as neither strategy alone is effective in the long term.

Benefits of a Balanced Approach:

- *Stable Learning:* Regular exploration prevents premature convergence, ensuring the agent refines its knowledge before committing to an action.
- *Effective Reward Accumulation:* With a balance between exploration and exploitation, the agent maximizes cumulative rewards over time.
- *Adaptability:* In dynamic environments, occasional exploratory actions enable the agent to detect changes and adapt to new reward distributions.

In summary, the exploration-exploitation trade-off ensures that an agent learns effectively while maximizing rewards. It prevents the agent from either wasting time on inefficient actions or prematurely converging to suboptimal solutions, leading to superior long-term performance.

4. Action-Value Methods (Problem 3)

An action-value method is a technique used to learn the best action value to effectively utilize exploitation. In reinforcement learning, agents make decisions based on these methods.

4.1 Action-Value Methods in Multi-Armed Bandit

In the context of the multi-armed bandit problem, the agent decides which arm to pull next based on the estimates of the value of each possible action (pulling an arm). These estimates are made using action-value methods.

4.2 General Steps in Action-Value Methods

1. Initial Value Estimates:

Since the agent initially has no knowledge of which arm has the best estimated reward value, a random choice is made while assigning an arbitrary initial value to each arm. In the project, we used two strategies for assigning initial values (Experiment 4):

- **Optimistic Initial Value:** $Q = 5, \varepsilon = 0$
- **Realistic Initial Value:** $Q = 0, \varepsilon = 0.1$

2. Action Selection:

The agent selects which arm to pull next based on the estimated action values. The action selection is made using methods such as epsilon-greedy or Upper Confidence Bound (UCB). We compared both epsilon-greedy and UCB methods in a stationary environment (Experiment 5).

- **Epsilon-greedy:** We balanced the trade-off between exploration and exploitation using the parameter ε , where the agent explored (selected random actions) with probability ε and exploited (selected the action with the highest estimated value) with probability $1 - \varepsilon$.
- **UCB:** UCB use confidence intervals (c) to balances exploration and exploitation systematically without randomness.

3. Updating Action-Value Estimates:

After each pull of an arm (action), the agent updates the estimated action-value based on the reward received. The goal is to incrementally adjust the estimates to converge to the true reward. We used two estimation methods—sample-average and exponential recency-weighted average—to update the action-value estimates in both stationary and non-stationary environments (Experiment 2 & 3).

- **Sample-average:** In the sample-average method, all rewards are given equal weights, which means that every reward contributes equally to the estimate of the expected reward. This approach works well in stationary environments where the underlying reward distribution does not change over time.

$$Q_{t+1} = Q_t + \frac{(\text{reward} - Q_t)}{\text{action count}}$$

- **Exponential recency-weighted average:** In non-stationary environments, the exponential recency-weighted average gives more weight to recent rewards to better adapt to changes over time.

$$Q_{t+1} = Q_t + \alpha \cdot (\text{reward} - Q_t)$$

5. Conclusion

- **Impact of Epsilon Value:** Increasing epsilon in the Epsilon-Greedy method led to faster convergence to better average rewards, highlighting the importance of exploration.
- **Stationary Bandit Performance:**
 - Sample averaging action estimation outperformed the exponentially weighted best estimation method, likely due to its stability in stationary environments.
 - Upper Confidence Bound (UCB) selection performed better than Epsilon-Greedy, demonstrating the advantage of confidence-based exploration.
- **Non-Stationary Bandit Performance:**
 - The exponentially weighted best estimation method outperformed sample averaging, as it adapts more effectively to changing reward distributions.
- **Greedy vs. Epsilon-Greedy:**
 - The purely greedy method ($\epsilon = 0$) resulted in suboptimal performance due to a lack of exploration.
 - The Epsilon-Greedy method ($\epsilon \neq 0$) performed better by ensuring periodic exploration of new actions.
- **Optimistic Initial Values:**
 - Encouraged more exploration, even with a low epsilon value in a greedy approach, reducing the risk of getting stuck in suboptimal choices.

References

[1] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA, USA: MIT Press, 2018. [Online]. Available:

<https://web.stanford.edu/class/psych209/Readings/SuttonBartoIPRLBook2ndEd.pdf>

Appendix

1. Contributions

Group Member	Contributions
Geo Raju (201855055)	• Experiment design • Multi-armed bandit implementation • Learning strategy comparison.

Anurag Gupta (201833580)	• Non-stationary bandit implementation • Greedy vs. ϵ-greedy performance analysis • Report writing (Methodology, Experiments & Result analysis)
Anito Anto (201853825)	• Experiment design • Refactored code • Report writing (Introduction, Methodology & Conclusions)
Nandana Puliyakkattil (201849311)	• Greedy vs. ϵ-greedy comparison • Exploration vs. exploitation study • Report writing (Exploration and Exploitation)
Nancy (201845219)	• Optimistic vs. realistic initialization comparison • Action-value selection method comparison • Report writing (Action-Value Methods)
Prachi Kadam (201837271)	• Non-stationary bandit implementation • Task and step function improvements • Report writing (Introduction)
Indudhar Chandrashekar (201824435)	• Stationary bandit analysis • Formatting, and code review.